# 3. Average velocity of cart's motion on an inclined plane – Lab guide

## Introduction

The experiment consists of releasing a cart from the top of an inclined plane and measuring its average velocity. The average velocity of the cart passing through the photogate, $v_{avg,exp}$, is the length of the picket fence on top of the cart $l$ $[m]$ (figure 3.1) divided by the average time it takes for the cart to pass through the photogate $\overline{\Delta t}$. The experiment will be repeated 150 times with the same initial conditions, and the statistical properties of the experimental system will be analyzed and compared to the statistical theory (Chapter 1.3 Statistics and distributions). Additionally, the experimental results will be compared with the theoretical model developed by the students using simple kinematical equations.

| In Shantou the value of the gravitational constant is [1]: |
|---|
| $\boldsymbol{g = 9.79127 \pm 0.000005\ [m/s^2]}$ |

The experimental setup is shown in Figure 3.1: The cart is released from the top of the inclined track towards a photogate that is placed at distance $s$ $[m]$. The acceleration of the cart is due to the gravity $g$ $[m/s^2]$. A picket fence with opaque line of length $l$ $[m]$ is placed on top of the cart. While the cart passes through the gate, the opaque line on the picket fence blocks the photogate. The photogate counts the blocking time, $\Delta t$ $[sec]$. Hereby, the photogate measures the time, $\Delta t$ $[sec]$, it takes for the cart to pass through it (please refer to the appendix for further explanation regarding the operation principle of the photogate).

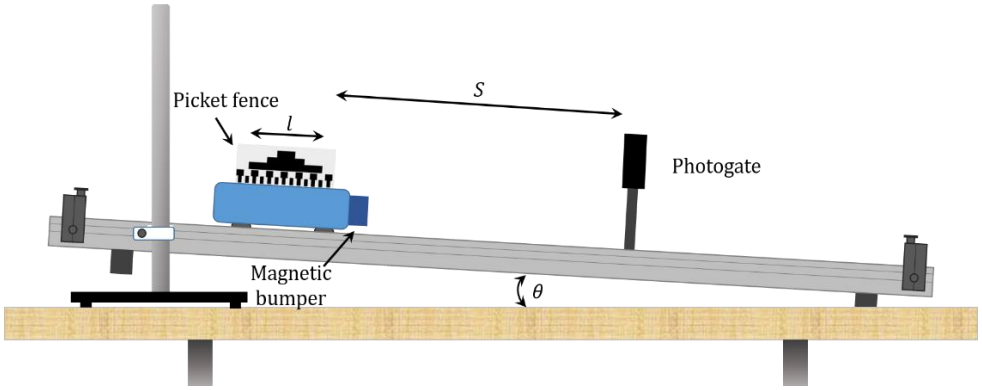



Figure 3.1 Experimental setup

Due to random error in the measurement of $\Delta t$, we will get a distribution of passing times. In the data analysis part, the 150 measured passing times, $\Delta t$, are plotted and compared to normal distribution. From the distribution of results, the mean passing time $\overline{\Delta t}$ and the error of measurement as standard error can be calculated.

# Preparation Questions

1. Explain why we need to repeat the experiment 150 times with the same initial conditions. Why one measurement is not enough? Why 10 measurements are not enough?
2. Explain why it is important that the measured $\Delta t$ will distribute normally? If the measured data distributes normally, what quantities can be derived from it?
3. Using Newton's laws, derive an analytical expression for the time it takes for the cart to pass through the photogate, $\Delta t\ [sec]$, under the assumption of negligible friction. Express your formula using experimental setup parameters: $l, s, \theta$ and the gravitational acceleration $g$.
4. Arrange the following data in increasing order of relative error:

$$g = 9.79127 \pm 0.000005\ \left[\frac{m}{s^2}\right]\ ;\ l = 0.200 \pm 0.001\ [m]\ ;$$

$$\theta = 2.52^{\circ} \pm 0.005\ ;\ s = 0.400 \pm 0.001\ [m]$$

   **What parameter contributes most significantly to the total error in $\Delta t$, and for which measured parameters the contribution for the error calculation in $\Delta t$ are negligible? Think how this helps you to calculate the error in the theoretical calculation of $\Delta t$.**
5. Write an expression for the error propagation of $\Delta t\ [sec]$ according to the expression you have derived in question 3. Assume errors of measurement of the experimental parameters: $\delta l, \delta s, \delta\theta$ and the gravitational acceleration constant $\delta g$ as in question 4.

# Experimental Procedure

a. Preparing the experimental system (figure 3.1)

**a.1** Ensure that the track is horizontal and stable (use the screws on the adjustable feet of the track fig, 3.2).

**a.2** Elevate one end of the track on a vertical rod using the pivot clamp (figure 3.1).

**a.3 Measure the inclination of the track using the digital level ruler. A Convenient choice (so that the cart won't hit the bumper stop when it reaches the end of the track) will be an inclination angle of less than $3^0$. Estimate the error in the angle measurement.**

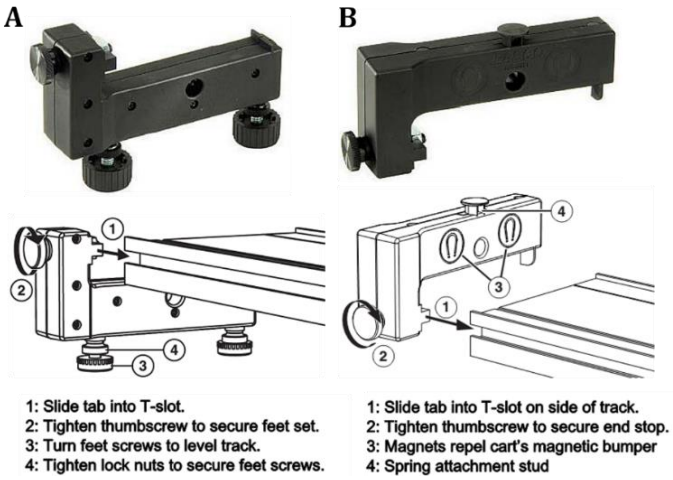



Figure 3.2 A. Adjustable feet. B. Track end Stop.

b. Connect the photogate to PASCO Capstone and load experiment setup file

**b.1** Plug in the photogate cable to the photogate on one side and to the *Airlink* digital adapter (figure 3.3) on the other side.

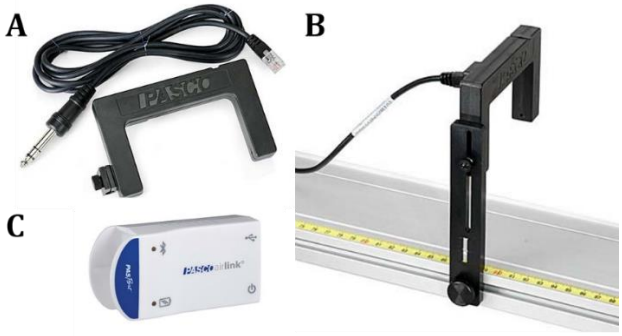



Figure 3.3 A. Photogate head. B. Photogate attached to the track. C. PASCO Airlink.

**b.2** Turn on *Airlink* and connect it to the computer using Bluetooth or using a USB cable.

**b.3** Start PASCO Capstone software. In tools palette (the most left tab). Open 'Hardware setup' and connect to *Airlink*. Pay attention to connect to your *Airlink* by the number written on it.

**b.4 Download the experiment setup file from Moodle: *AV_exp_setup*. If needed, change file type from "*.zip*" to file type "*.cap*". Save the file in your working folder for the current experiment.**

**b.5** In PASCO Capstone, click on 'file' and open the experiment setup file '*AV_exp_setup.cap*'.

c. Preliminary measurements

**c.1** Attach the picket fence on top of the cart (figure 3.1).

**c.2 Adjust the height of the photogate so that the longest bar on the picket fence will block the photogate detector when the cart passes through the gate. When the cart is within the gate, the red light of the gate will shine.**

**c.3** Choose a release distance of the cart from the gate.

**c.4 To determine $s$ and $l$, pass the cart slowly with your hand through the gate and notice how far the cart travels from the release point until the gate is being blocked (this distance is $s$). Secondly, notice the distance the car travels while the gate is blocked (this distance is $l$). Estimate the error in each measured quantity.**

**c.5** Preliminary experiment: Release the cart from the same distance to the gate several times. Be sure that the initial conditions for each release are identical (from rest and from the same distance from the photogate). Observe the result of each measurement shown in the display and check that there are no abrupt changes from one measurement to the other. Think about the possible sources of random errors and try to reduce them. Pay attention to any factor that can interfere with the stability of the movement of the cart.

d. Average passing time measurement

**d.1** Clear the data collected in the preliminary experiment.

**d.2 Release the cart $n = 150$ times with the same initial conditions (from rest and from the same distance from the photogate). If you perform a wrong measurement (for example, if the cart was released with some initial push, or from a different initial place), then take account of the number corresponding to such measurement to be able to identify it at the end of the experiment and erase it from the data.**

**d.3** Select the data in the table and copy it. Open notepad and paste the data there. Delete the first line so there will be only numbers written in the file and save it as a *txt* file.

# Data analysis

Part A: Experiment log and theoretical model

1. Open a new python script and write down the experiment log: Write in all the directly measured parameters and their uncertainties. Using comments, include the units of every measured parameter.
2. **Calculate the theoretical value of cart's velocity while passing through the photogate, $\Delta v$, according to the theoretical model you have developed. Hint: See preparation question 3. Verify with your lab instructor that your theoretical model is correct.**
3. Calculate the uncertainty in $\boldsymbol{\Delta v}$ using error propagation. **Show your calculation process in the lab report appendix.** Hint: See preparation questions 4 and 5.

Part B: Outcome of a repeated experiment and comparison to the theory

4. Open Python and import the measured data *txt* file. Hint: Save the script and the data file in the same folder and use the function "loadtxt".
5. **Plot a histogram for the measured values of $\Delta t$ using matplotlib function 'hist'. Select an appropriate number of bins for the histogram. What considerations you made for your selection? Create another histogram for the average velocities. Make sure the numerical values you get make sense.**
   **Hint: Use the link on Moodle "How to plot histogram and normal distribution".**
6. Remove from the data any outliers. Show your process in the lab report appendix. Hint: Read again the "1.3 statistics and distribution" chapter.
7. **What is the outcome of the experiment? To find it, calculate the average velocity of the cart through the photogate: $\overline{\Delta v}$, the standard deviation $s$ and the standard error $s_m$. Hint: Read the first two pages in the "1.3 statistics and distribution" chapter.**
8. Plot a histogram of your data without the outliers. Add a theoretical line of a normal distribution using $\overline{\Delta v}$, and $s$ you have calculated. Add this plot to the lab report and discuss about the comparison between the experimental histogram and the normal distribution. Hint: Read again the "1.3 statistics and distribution" chapter about normal distribution.
9. **Compare between the theoretical and the experimental values $\Delta v$. Answer in your lab report: Does the experiment agree with the theory? If they agree, state your conclusion from the experiment. If they disagree, explain the errors leading to the disagreement. Hint: Find how to compare between measured parameters in the "1.3 statistics and distribution" chapter.**

Part C: Statistical theory

10. **Question: Assuming only random errors, why are the results of a repeated measurement distributed normally? Short answer: The central limit theorem (see "1.3 statistics and distribution")**

    To understand the concept of the central limit theorem, perform steps 11.1-11.3 two times: First work with sample of 20 measurements ($\boldsymbol{N = 20}$), then repeat for 60 measurements ($\boldsymbol{N = 60}$):

    11.1 Calculate the mean velocity while passing of the cart through the photogate: $\overline{\Delta v}_N$, the standard deviation $s_N$ and the standard error $s_{m,N}$.

    11.2 Plot a histogram of the sample.

    11.3 To the histogram plot, add a theoretical line of the normal distribution, using $\overline{\Delta v}_N$, and $s_N$.

    **Discuss the characteristic of the distribution you got from the measurements for the different sample sizes ($\boldsymbol{N = 20}$, $\boldsymbol{N = 60}$ and $\boldsymbol{N = 150}$): and compare it to the theoretical normal distribution. Answer in your lab report: Why do we expect the results to distribute normally? Do the different samples have the properties of normal distribution?**

11. **At the end of work, upload your work files (data file and Python script) to the course's site under the Average velocity section. Without these files there will not be grade for the lab work!**

## References

[1] Adams, B. (06.02.2014), Local Acceleration of Gravity,
http://www.wolframalpha.com/widgets/view.jsp?id=e856809e0d522d3153e2e7e8ec263bf2

[2] Equipment images (figure 3.2.2 and 3.2.3) were taken from PASCO site:
https://www.pasco.com

[3] More information on the photogate
https://www.pasco.com/prodCompare/smart-gate/index.cfm